

Project Task: E-Commerce Customer Segmentation Using K-Means Clustering

Project Overview

Build a Customer Segmentation System using K-Means Clustering to group customers based on purchasing behavior and generate business insights.

Objective

Develop a clustering pipeline to segment customers, determine the optimal number of clusters, visualize the clusters, analyze each segment, and provide business recommendations.

Dataset Requirements

Customer ID
Age
Gender
Annual Income
Average Monthly Spending
Total Orders
Average Order Value
Purchase Frequency
Discount Usage
Days Since Last Purchase
Loyalty Membership Status

Note: No target variable is available because K-Means is an Unsupervised Learning algorithm.

Functional Requirements

Module 1: Exploratory Data Analysis

- Dataset overview
- Missing value analysis
- Duplicate analysis
- Statistical summary
- Correlation analysis
- Distribution plots
- Correlation Heatmap

Module 2: Data Preprocessing

- Handle missing values
- Remove unnecessary columns
- Encode categorical variables
- Apply StandardScaler or MinMaxScaler
- Select relevant features

Module 3: Find Optimal K

- Apply Elbow Method
- Plot WCSS graph
- Determine optimal K
- Validate using Silhouette Score

Module 4: K-Means Model

- Train K-Means
- Assign cluster labels
- Add cluster column
- Display customer count in each cluster

Module 5: Cluster Visualization

- Income vs Spending
- Purchase Frequency vs Average Order Value
- Plot Centroids
- Cluster Distribution